

CHEA ROTH

DATA SCIENCE AND MACHINE LEARNING

Engineering Degree in Apply Mathematics and Statistics

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SUMMARY

Data Scientist with an **Engineering Degree in Data Science** from the **Institute of Technology of Cambodia (ITC)**. Skilled in Machine Learning, Time Series Forecasting, and Data Visualization, with hands-on experience developing predictive models, hybrid ML frameworks, and real-world analytical projects. Strong foundation in statistics, programming, and research, with a proven ability to turn data into actionable insights for decision-making and policy planning.

TECHNICAL SKILLS

Programming & Tool

- Python, SQL, HTML/CSS
- Git, Jupyter Notebook, Google Colab, VS Code, Anaconda
- Excel, Google Sheets

Machine Learning & AI

- Regression, Classification, Random Forest, XGBoost, SVR
- LSTM, SARIMAX, Hybrid ML Models
- Scikit-learn, TensorFlow, Keras, LightGBM, CatBoost

Time Series & Forecasting

- ARIMA, SARIMA, LSTM, Prophet
- Exponential Smoothing, Statsmodels, PyFlux
- Trend & Seasonality Analysis

Data Analysis & Visualization

- EDA, Feature Engineering, Data Cleaning
- Matplotlib, Seaborn, Plotly, Tableau, Power BI
- Pandas, NumPy

Natural Language Processing (NLP)

- SentencePiece, Tokenization, Corpus Creation, LoRA Fine-Tuning
- Hugging Face Transformers, NLTK, SpaCy, Gensim

Database Management

- MySQL, PostgreSQL, SQLite, Microsoft Access, MongoDB (basic)
- SQL Queries, ERD Design, Relational Modeling

Other Strengths

- Statistics, Research Writing, Communication, Data Storytelling
- Problem Solving, Critical Thinking, Business Insight

PROFESSIONAL EXPERIENCE

Ministry of Planning, Cambodia | *Data Scientist*

Feb 2025 – Oct 2025

- Developed a hybrid inflation forecasting model (XGBoost + LSTM + SARIMAX), improving accuracy of national economic projections.
- Built and maintained data pipelines for macroeconomic indicators, including cleaning, feature engineering, and stationarity testing.
- Conducted analysis of global (oil, gold) and domestic sector factors affecting inflation.
- Created interactive dashboards and reports to support policy-making and economic planning.
- Collaborated with economists and policymakers to translate complex data into actionable insights.

Sunrise Institute | *Instructor – Data Science*

Jan 2024 – Feb 2025

- Taught EDA, statistics, machine learning, forecasting, and data visualization with a focus on practical Python and Power BI projects.
- Guided students in building dashboards, performing analysis, and deriving actionable insights from real datasets.
- Designed hands-on exercises and mini-projects to strengthen students’ analytical, problem-solving, and coding skills.
- Mentored students on data storytelling and presentation, preparing them to communicate insights effectively to technical and non-technical audiences.
- Supported students in applying ML models and statistical methods to solve real-world business and research problems.

SELECTED PROJECT

Laptop Price Prediction

Role: Data Analyst & Model Developer

Tools: Python, Pandas, Scikit-learn, BeautifulSoup, Matplotlib, Seaborn

- Performed data cleaning, exploratory data analysis (EDA), and feature engineering.
- Applied multiple regression models including Linear Regression, Random Forest, and Support Vector Regressor (SVR).
- Achieved high prediction accuracy with the Support Vector Regressor model.

Forecasting Tourism in Cambodia

Role: Data Analyst / Time Series Forecaster

Tools & Skills: Python (Pandas, NumPy, Matplotlib), ARIMA, SARIMA, Prophet, LSTM, Time Series Modeling

- Collected and preprocessed tourism data for time series analysis.
- Evaluated model performance using Mean Squared Error (MSE) and residual diagnostics.
- Identified seasonal patterns and recovery trends post-COVID-19.
- Recommended data-driven insights for tourism recovery planning and policy-making.

Hotel Reservation System Database

Role: Database Designer & Developer

Tools & Skills: MySQL, SQL, ERD Design, Relational Database Modeling

- Designed and implemented a database system to manage hotel operations such as room booking, client management, and staff scheduling.
- Built ER diagrams and relational schemas for key entities Client, Staff, Room, Service, Floor, Booking.
- Applied data normalization, primary and foreign key constraints to ensure data integrity.
- Supported receptionist workflows including booking verification and payment confirmation.

Crop Yield Prediction & Recommendation

Role: Data Analyst

Tools & Skills: Python, Pandas, NumPy, Scikit-learn, XGBoost, Matplotlib, Seaborn, Jupyter Notebook

- Collected and cleaned agricultural datasets.
- Built machine learning models (Random Forest, XGBoost) to predict crop yield.
- Engineered key features from soil and weather data.
- Evaluated model performance using R², MAE, and RMSE.
- Developed a recommendation system for suitable crop selection.

EDUCATION

Institute of Technology of Cambodia (ITC)

2020 – 2025

Engineering Degree in Data Science

Major in Applied Mathematics and Statistics

Thesis: Analysis and Forecasting of Inflation in Cambodia

Aii Language Center

2019 – 2022

English

French (Basic)

REFERENCE

PHAUK SOKKHEY, Ph.D

Position: Deputy head of department of applied mathematics and statistics (AMS)
(អនុប្រធានដេតីម៉ង់គណិតវិទ្យាអនុវត្ត និងស្ថិតិ) and Head of Master Program of Data Science

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Mr. Pakrina Long

Position: IT Lectureer in DATA Science and Institute of Technology Cambodia

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